Quadratic remainder sums for two primes congruent to three modulo four

Principia Math In Progress

29 September 2026

Abstract

We prove Chamberland–Dilcher Conjecture 7.2 for all positive exponents. The proof expands the distribution of unit squares in quadratic characters and partitions the original sum according to the two prime valuations. This produces the conjectured class-number coefficients, including the exceptional normalization at the prime three.

Task 1: the original conjecture

Let p, q be distinct primes congruent to 3 modulo 4, let $a, b \geq 1$, and set $n = p^a q^b$. Define

$$S_n = \frac{1}{n} \sum_{k=1}^{n-1} (k^2 \bmod n) = \sum_{k=0}^{n-1} \left\{ \frac{k^2}{n} \right\}, \qquad G_t(x) = \frac{x^t - 1}{x - 1} \quad (t \geq 0).$$

Here remainders are least nonnegative remainders and braces denote fractional parts. Write

$$u = \lfloor a/2 \rfloor, \quad v = \lfloor b/2 \rfloor, \quad A = \lceil a/2 \rceil, \quad B = \lceil b/2 \rceil, \qquad H_r = h^*(-r).$$

For a prime $r \equiv 3 \pmod 4$, H_r is the class number of $\mathbb{Q}(\sqrt{-r})$ when $r > 3$, while $H_3 = 1/3$.

Theorem 1. *With this notation,*

$$\begin{aligned} S_n = \frac{n - p^u q^v}{2} & \\ & - G_A(p) \left(G_{v+1}(q) + \left(\frac{p}{q} \right) G_v(q) \right) H_p \\ & - G_B(q) \left(G_{u+1}(p) + \left(\frac{q}{p} \right) G_u(p) \right) H_q. \end{aligned}$$

Since $A = \lfloor (a+1)/2 \rfloor$ and $B = \lfloor (b+1)/2 \rfloor$, this is exactly Conjecture 7.2, Equation (7.2), in [1, p. 21]. There is no restriction on the sizes or parities of the positive exponents.

Unit-square averages

Let $\chi_r(x) = \left(\frac{x}{r}\right)$. We use the classical finite class-number formula

$$\frac{1}{r} \sum_{x=1}^{r-1} x \chi_r(x) = -H_r. \tag{1}$$

This is Dirichlet's formula as stated in [1, Theorem 4.1 and Corollary 4.3]. At $r = 3$ the sum is $(1-2)/3 = -1/3$, which explains the stated normalization.

For odd $M > 1$ and c coprime to M, put

$$U_M(c) = \sum_{\substack{x \bmod M \\ (x,M)=1}} \left\{ \frac{cx^2}{M} \right\}.$$

For $e, f \geq 1$ the following formulas hold:

$$U_{p^e}(c) = \frac{\varphi(p^e)}{2} - \chi_p(c) H_p, \tag{2}$$

$$U_{p^e q^f}(1) = \frac{\varphi(p^e q^f)}{2} - (1 - \chi_p(q)) H_p - (1 - \chi_q(p)) H_q. \tag{3}$$

We give the details so that their application to nonunit terms is explicit.

For an odd prime r, a unit residue y modulo r^e has $1 + \chi_r(y)$ square roots. Indeed, modulo r it has zero or two roots, and each root lifts uniquely at each step: solving $(x + tr^j)^2 \equiv y \pmod{r^{j+1}}$ is a linear equation for t with nonzero coefficient $2x$ modulo r. Consequently the number of roots of $cx^2 = y \pmod{p^e}$ is $1 + \chi_p(c)\chi_p(y)$. Summing y/p^e over unit residues y gives (2): the principal sum is $\varphi(p^e)/2$ by pairing y with $p^e - y$, and the character sum is $-H_p$ by splitting into periods of length p in (1).

For $M = p^e q^f$, the Chinese remainder theorem gives the root count

$$(1 + \chi_p(y))(1 + \chi_q(y)) \qquad ((y, M) = 1).$$

The principal term again sums to $\varphi(M)/2$. The product character $\chi_p \chi_q$ is even and nonprincipal, so its weighted sum is zero: pairing y with $M - y$ reduces it to half its unweighted sum, which vanishes. Finally, by periodicity and (1),

$$\frac{1}{M} \sum_{y=1}^{M} y \chi_p(y) = -H_p, \qquad \frac{1}{M} \sum_{\substack{1 \leq y \leq M \\ q \mid y}} y \chi_p(y) = -\chi_p(q) H_p.$$

Subtracting excludes multiples of q; multiples of p already contribute zero. Interchanging p and q proves (3).

Partition by valuations

For $0 \leq k < n$, put

$$i = \min(v_p(k), a), \qquad j = \min(v_q(k), b),$$

where the valuations of zero are taken to be infinite. In a fixed class (i, j), write $g = p^i q^j$, $m = n/g$, and $k = gx$. Then x runs once through the unit residues modulo m (with the sole zero residue when $m = 1$).

Call the p valuation active when $2i < a$, equivalently $0 \leq i < A$; define active q valuations analogously. If both are active, set $M = p^{a-2i} q^{b-2j}$. Reduction from units modulo m to units modulo M has constant fiber size $p^i q^j$, and $k^2/n = x^2/M$. Thus this class contributes

$$p^i q^j U_{p^{a-2i} q^{b-2j}}(1). \tag{4}$$

If only the p valuation is active, then $0 \le i < A$ and $B \le j \le b$. Now

$$\frac{k^2}{n} = \frac{q^{2j-b}x^2}{p^{a-2i}},$$

and reduction to units modulo p^{a-2i} has constant fiber size $p^i\varphi(q^{b-j})$, with $\varphi(1) = 1$. The contribution is

$$p^i\varphi(q^{b-j})U_{p^{a-2i}}(q^{2j-b}). \tag{5}$$

The formula with p and q interchanged treats the third case. If neither valuation is active, then $n \mid k^2$ and the contribution is zero. These four cases exhaust every residue k without overlap.

Collecting the coefficients

In each nonzero class, the principal term in (2) or (3), multiplied by its fiber size, is half the number of residues in that class. There are exactly p^uq^v residues satisfying $n \mid k^2$: they are the multiples of p^Aq^B modulo n. The total principal contribution is therefore $(n - p^uq^v)/2$.

Put $s = \chi_p(q) \in \{-1, 1\}$. From (4), the coefficient of $-H_p$ in the classes with both valuations active is

$$G_A(p)(1 - s)G_B(q).$$

In (5), the twisting character has constant value $\chi_p(q^{2j-b}) = s^b$. Since

$$\sum_{j=B}^{b} \varphi(q^{b-j}) = \sum_{t=0}^{v} \varphi(q^t) = q^v,$$

these classes contribute $G_A(p)s^bq^v$ to the same coefficient. The classes with only q active have no H_p term. Hence its full coefficient is

$$C_p = G_A(p)\big((1 - s)G_B(q) + s^bq^v\big).$$

Checking $b = 2v$ and $b = 2v + 1$ separately, using $G_{v+1}(q) = G_v(q) + q^v$, gives in both cases

$$(1 - s)G_B(q) + s^bq^v = G_{v+1}(q) - sG_v(q).$$

Quadratic reciprocity for $p \equiv q \equiv 3 \pmod 4$ gives $-s = \left(\frac{p}{q}\right)$. This yields the first class-number coefficient in Theorem 1. Interchanging p, a with q, b gives the second. Together with the principal contribution, this proves the theorem.

Tasks 2–5: follow-up questions

2. Derive a corresponding explicit valuation formula for three or more distinct odd prime factors, tracking every odd quadratic character.

3. Extend the unit-square calculation to moduli divisible by arbitrary powers of two, with explicit conductor and exceptional-unit corrections.

4. Prove uniform error bounds for $S_n - n/2$ as n varies through integers with a fixed number of distinct prime factors.

5. Determine exact formulas for higher moments of the normalized remainders $\{k^2/n\}$ and identify the Dirichlet special values that occur.

Verification. The accompanying integer/rational verifier evaluates the original sum, enumerates reduced binary quadratic forms to check class numbers independently, and checks the coefficient simplifications. Finite tests corroborate the written all-exponent proof. Review is by the author-agent; no independent peer review is claimed. Source and bounded-search limits are recorded in the literature audit.

References

[1] M. Chamberland and K. Dilcher, *Sums of the floor function related to class numbers of imaginary quadratic fields*, arXiv:2510.04387v1 (2025). The complete current 23-page manuscript was read on 29 September 2026.

[2] MathDB, problem 370786, live record checked 29 September 2026.